

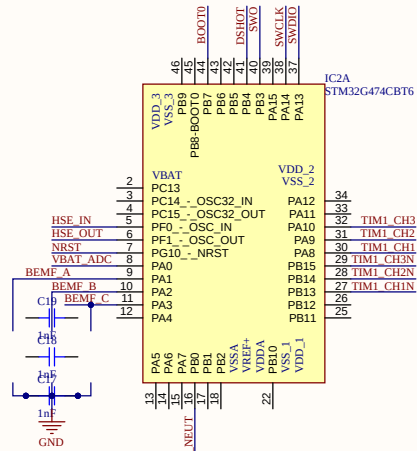
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Wiz ESC

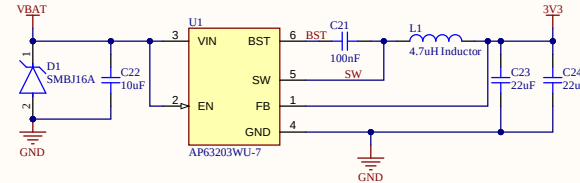
-Custom sensorless electronic speed controller
-Nomially takes a 3S battery but can take up to a 4S

REVNO.	NO.	REV.	DATE	APPROVED
1	1			

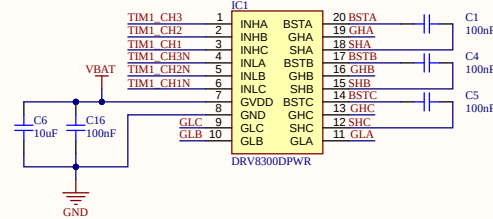
STM32G4 MCU



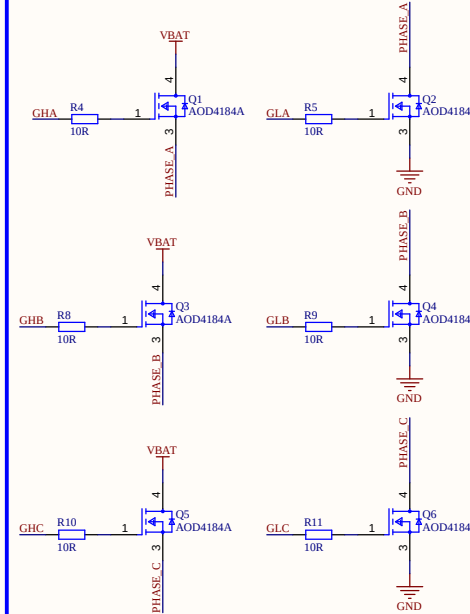
VBAT to 3V3 and protection



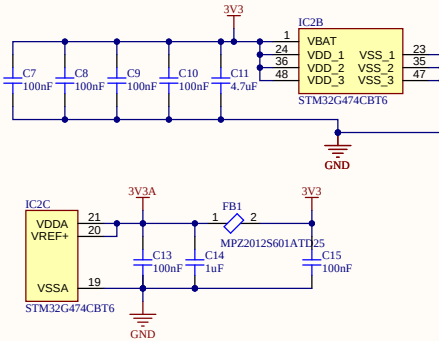
Gate Driver and circuitry



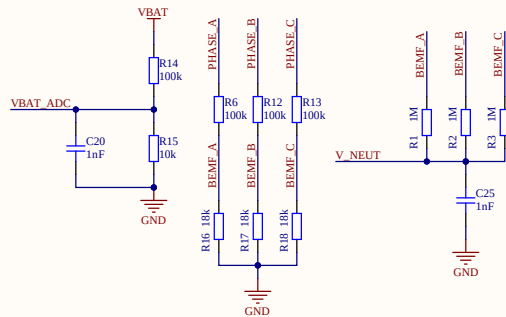
Mosfets and phase outputs



STM decoupling and analog

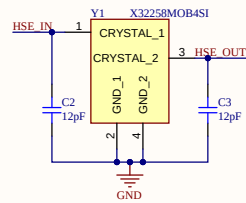


Voltage sensing for VBAT, back emf and neutral point

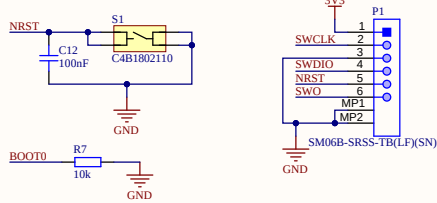


Notes:
-Values for resistors and capacitors have largely been chose based of datasheet recommendations.
-Except for C_GVDD, C_BSTx and R_GHx/GLx on the gate driver/mosfet gates.
-Voltage divider resistances have been decided on to ensure no damage is done to the MCU ADC pins and also allow for relatively accurate readings.
-Other valeus not mentioned in any datasheets, e.g. 1nF decoupling on the ADC lines has been chosen out of respect for the nature of their purpose.
-Generally parts chosen for simplicity, good documentation and overall fitting the needs.

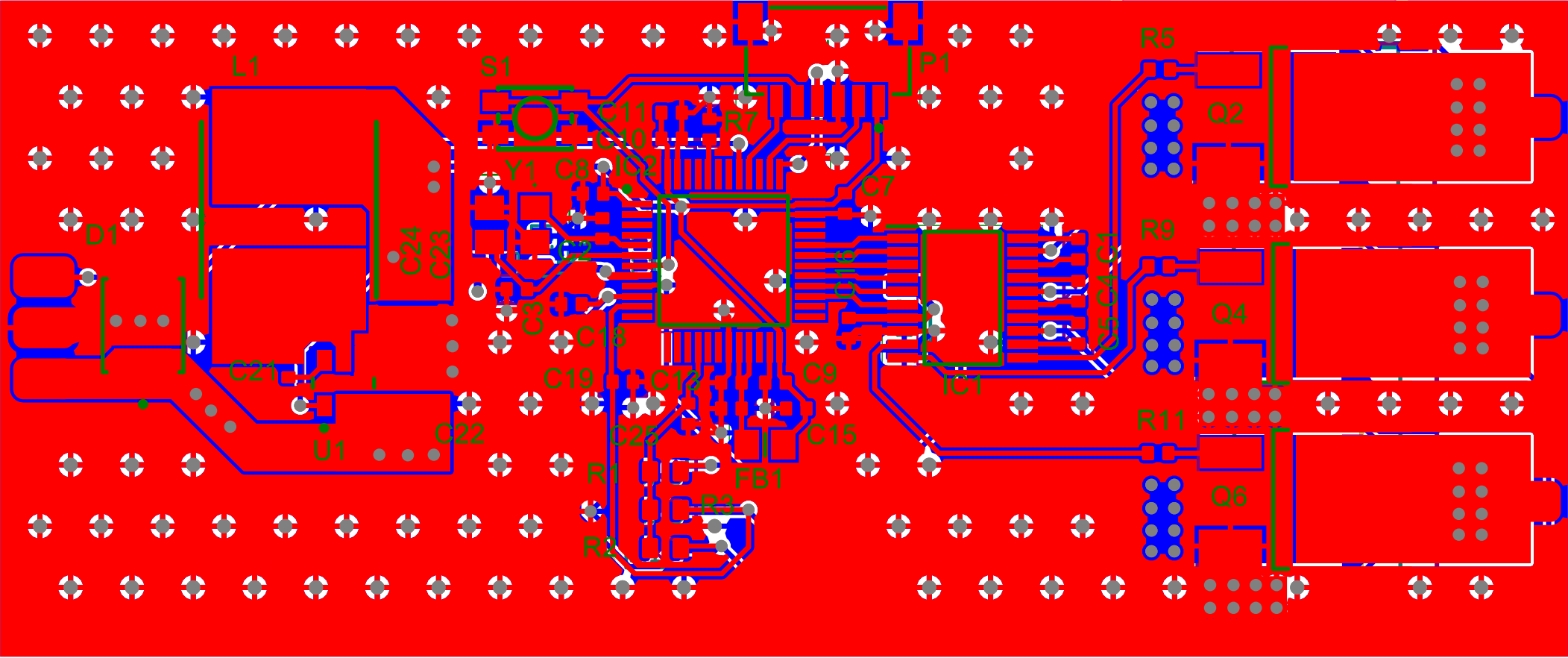
8MHz Crystal for STM



Boot, Reset and SWD for STM



APPROVALS	DATE	PROJECT	Altium
ENG: Barney		PROJECT REVISION:	DOCUMENT REVISION:
DSN: Barney		TITLE	Wiz ESC
CHK: Barney		SCALE	FILENAME
BOM:		SIZE	CAGE CODE
ASSY DWG:		DWG NO.	REV
FAB DWG:		SCALE	FILENAME
PCB DWG:		SHEET	OF



Name	Designator	Footprint	LCSC	Quantity	Manufacturer	Manufacturer Part Number
100nF	C1, C4, C5, C7, C8, C9, C10, C12, C13, C15, C16, C21	CAP_0402	C960916, C100072	12	Samsung Electro-Mechanics	CL05B104KB5NNNC, CL05A104KA5NNNC
12pF	C2, C3	CAP_0402	C26406	2	Samsung Electro-Mechanics	CL05C120JB5NNNC
10uF	C6, C22	CAP_1206	C2803389	2	Samsung Electro-Mechanics	CL31Y106KBKVPNE
4.7uF	C11	CAP_0402	C368809	1	Samsung Electro-Mechanics	CL05A475KP5NRNC
1uF	C14	CAP_0402	C52923	1	Samsung Electro-Mechanics	CL05A105KA5NQNC
1nF	C17, C18, C19, C20, C25	CAP_0402	C14442	5	Samsung Electro-Mechanics	CL05B102KB5NNNC
22uF	C23, C24	CAP_0402	C2762594	2	Samsung Electro-Mechanics	CL10A226MO7JZNC
SMBJ16A	D1	FP-DO-214AA-MFG	C353386	1	Jingdao Microelectronics	SMBJ16A
MPZ2012S601ATD25	FB1	BEADC2112X105N	C81034	1	TDK	MPZ2012S601ATD25
DRV8300DPWR	IC1	SOP65P640X120-20N	C3036056	1	Texas Instruments	DRV8300DPWR
STM32G474CBT6	IC2	QFP50P900X900X160-48N	C730123	1	STMicroelectronics	STM32G474CBT6TR
4.7uH Inductor	L1	SMD 8x8 Inductor	C266436	1	Shun Xiang Nuo	SMNFR8040-4R7MT
SM06B-SRSS-TB(LF)(SN)	P1	JST-SM06B-SRSS-TB_V	C160405	1	JST	SM06B-SRSS-TB(LF)(SN)
AOD4184A	Q1, Q2, Q3, Q4, Q5, Q6	FP-TO252-MFG	[NoParam], C99124	6	[NoParam], AOS	[NoParam], AOD4184A
1M	R1, R2, R3	RES_0603	C2907003	3	FOJAN	FRC0603F1004TS
10R	R4, R5, R8, R9, R10, R11	RES_0402	C2906862	6	FOJAN	FRC0402F10R0TS
100k	R6, R12, R13, R14	RES_0603	C2907088	4	FOJAN	FRC0603J104 TS
10k	R7	RES_0402	C2906861	1	FOJAN	FRC0402F1002TS
10k	R15	RES_0603	C2906982	1	FOJAN	FRC0603F1002TS
18k	R16, R17, R18	RES_0603	C2907001	3	FOJAN	FRC0603F1802TS
C4B1802110	S1	SW-SMD_4P-L3.5-W3.0-P1.50-LS4.7	C49393608	1	SAMZO	C4B1802110
AP63203WU-7	U1	FP-TSOT26-MFG	C780769	1	Diodes Inc.	AP63203WU-7
X32258MOB4SI	Y1	X32258MOB4SI	C2682775	1	YXC	X32258MOB4SI